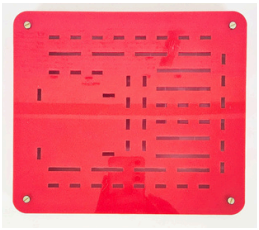
ScienceUtsav®
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# Smart Switching and Door Sensor Automation

## Session 4

# STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


2


3


4


5

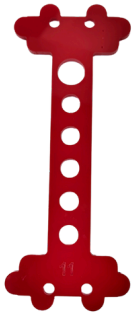

6


7


8


9


10


11


1

Spacers

2

L Clamp

3

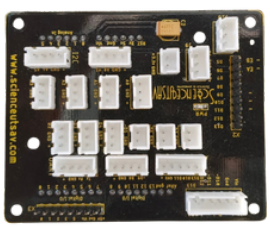
Spacer Screws

4

Nuts

5

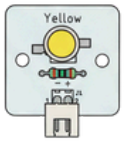
Screw

6

Arduino With Shield

7

Arduino cable

# ELECTRONIC COMPONENTS



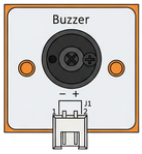

**1 Watt LED PCB**

1 Watt LED is a high-brightness light-emitting diode that produces much stronger light compared to normal small LEDs.



**DIP LED**

A DIP LED (Dual In-line Package LED) is a type of LED with two long metal pins used for easy connection on breadboards and PCB circuits.




**BUZZER PCB**

A buzzer is an electronic component that generates sound when electric current flows through it. It is commonly utilized in alarms, alerts, and notification systems to provide auditory signals.



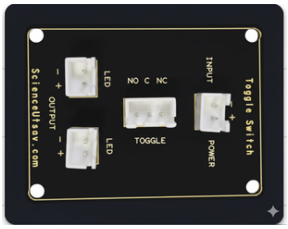
**limit switch**

A limit switch is a mechanical switch that gets activated when an object touches or presses it. It is used to detect movement, position, or opening/closing actions in machines and automation projects.



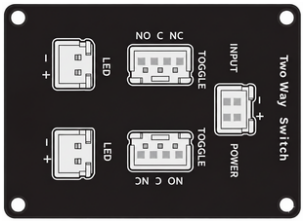
**toggle switch**

A toggle switch is a switch that stays in its selected position (ON or OFF) until it is manually changed. It is commonly used to control circuits and devices by switching the power flow permanently until toggled again.




**toggle switch PCB**

A toggle switch PCB is a small circuit board made to mount and connect a toggle switch neatly in an electronic circuit.

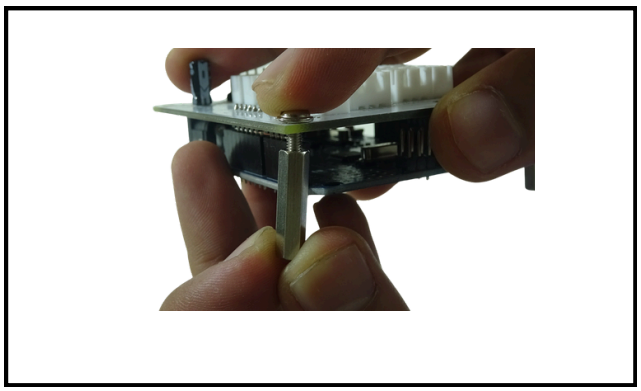


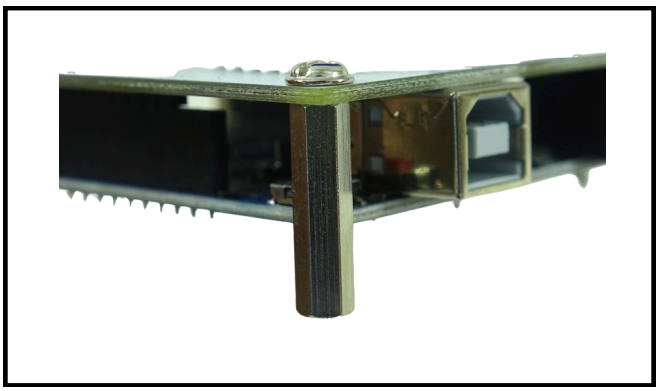

**two-way switch PCB**

A two-way switch PCB is a circuit board designed to connect two switches that can control the same light or device from two different locations.

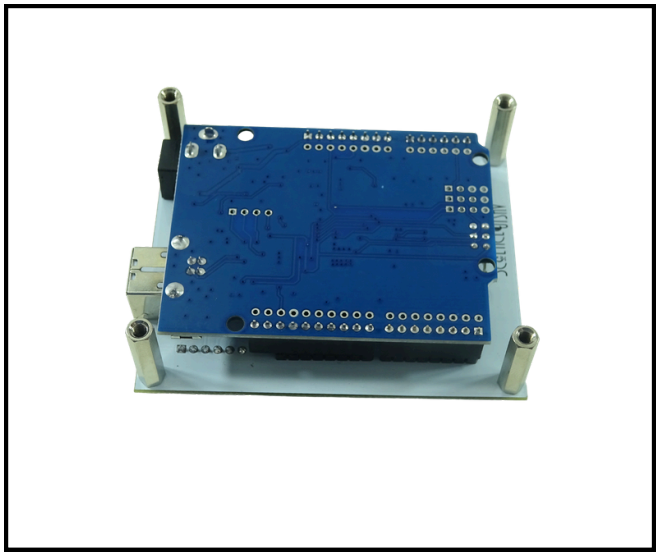
# Procedure :

**Step 1 :** Fit the spacer to Arduino Shield as shown below using Spacer screw

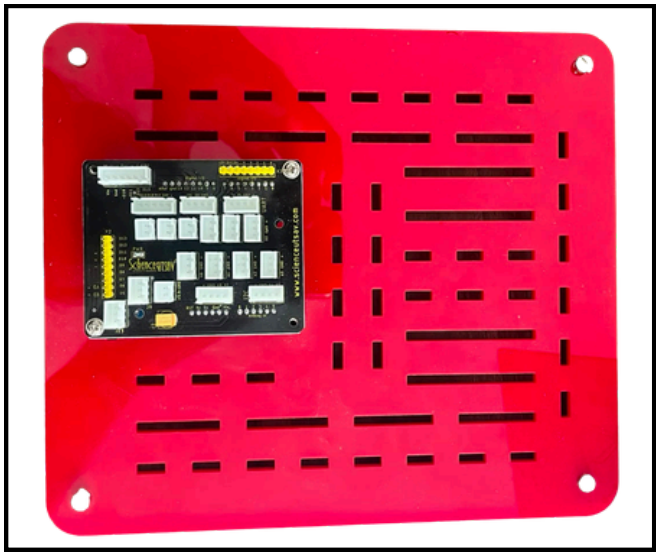




**Step 2 :** Repeat the step 1 for all 4 corners of Arduino Shield



**Step 3 :** Place the Arduino Shield to the base board as shown below

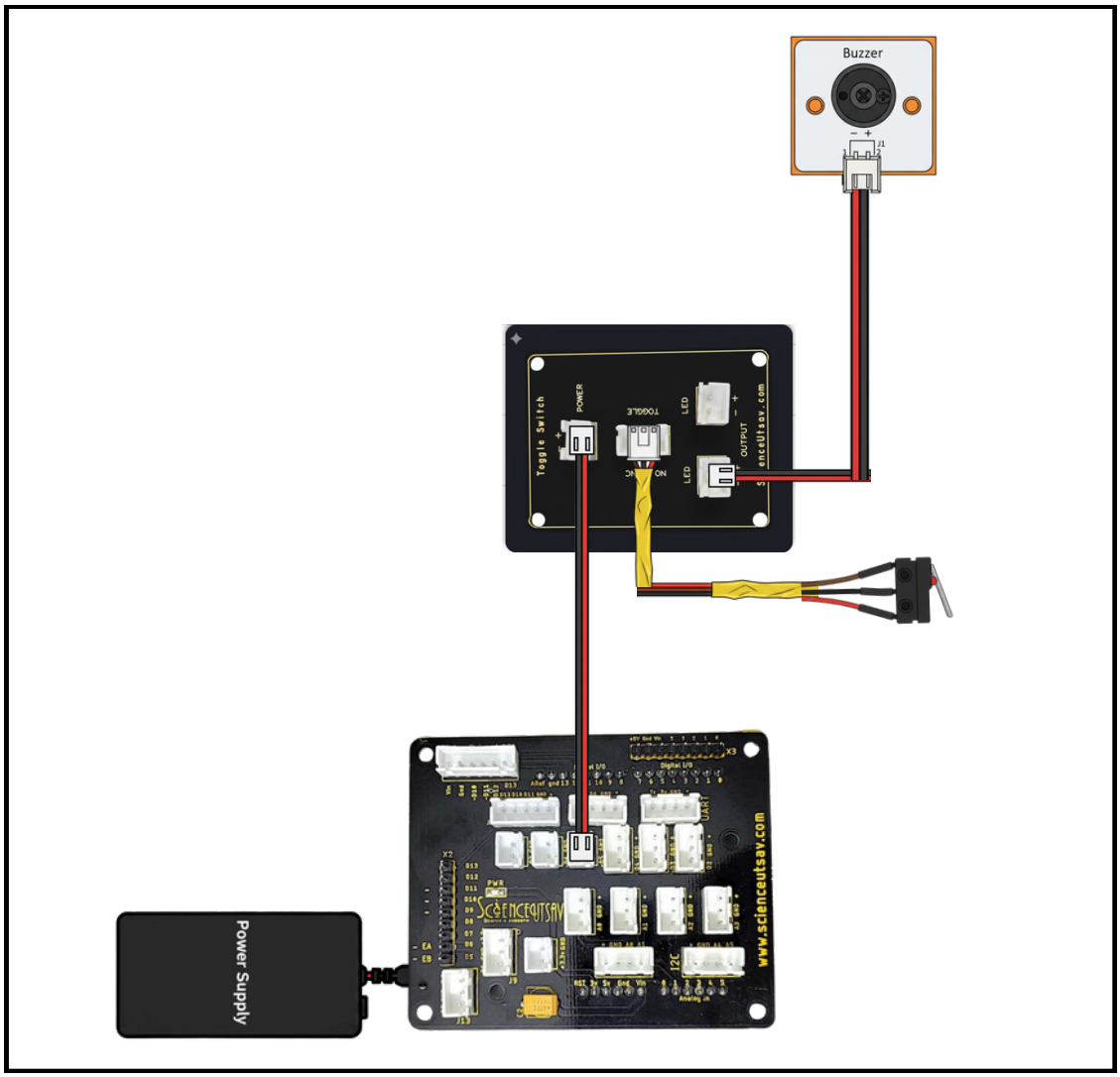


**Step 4 :** Fix it with the spacer screw from the bottom as shown below



# BEGINNER PROJECT

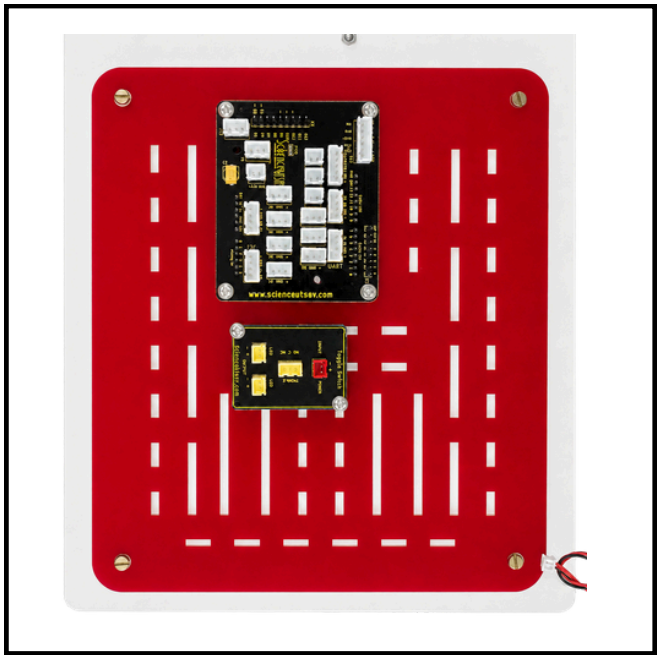
The circuit diagram of the project is given below



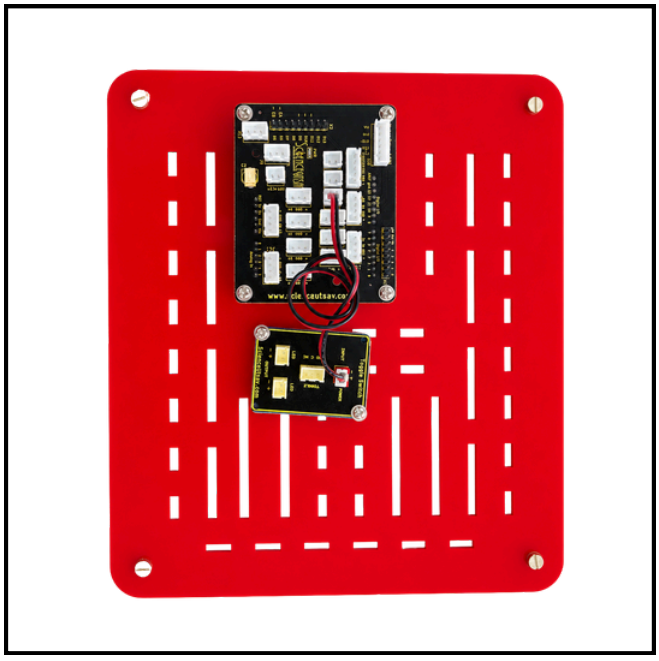

## CONNECTION:

1. TOGGLE SWITCH PCB IN D5
2. TOGGLE SWITCH PCB- LIMIT SWITCH, BUZZER

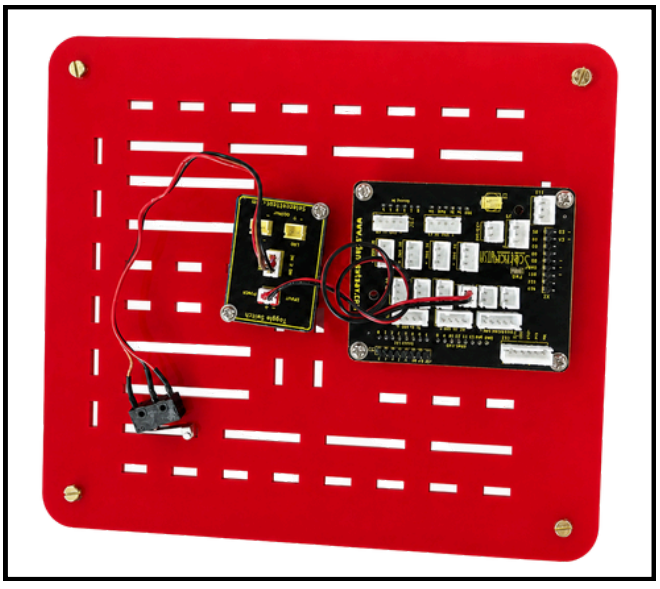
**Step 1:** Secure the Toggle Switch PCB to the Pegboard



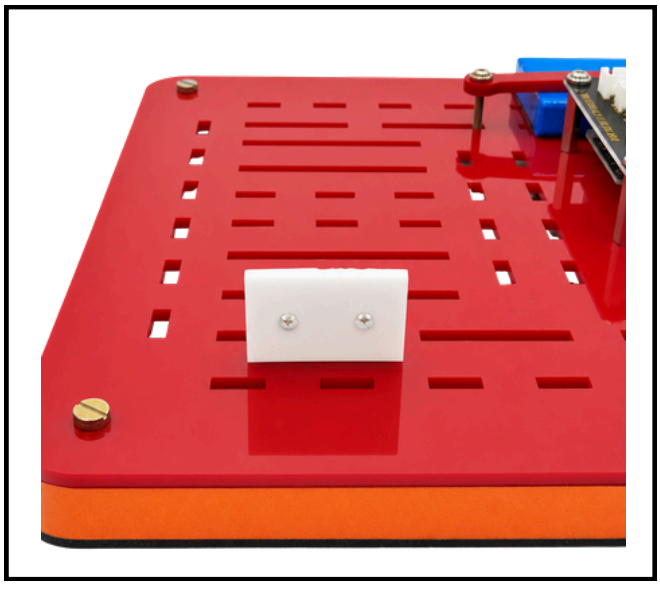
**Step 2:** Connect the 2-pin RMC to the D5 and to the toggle switch PCB.



**Step 3:** Connect the limit switch to the PCB as illustrated below.

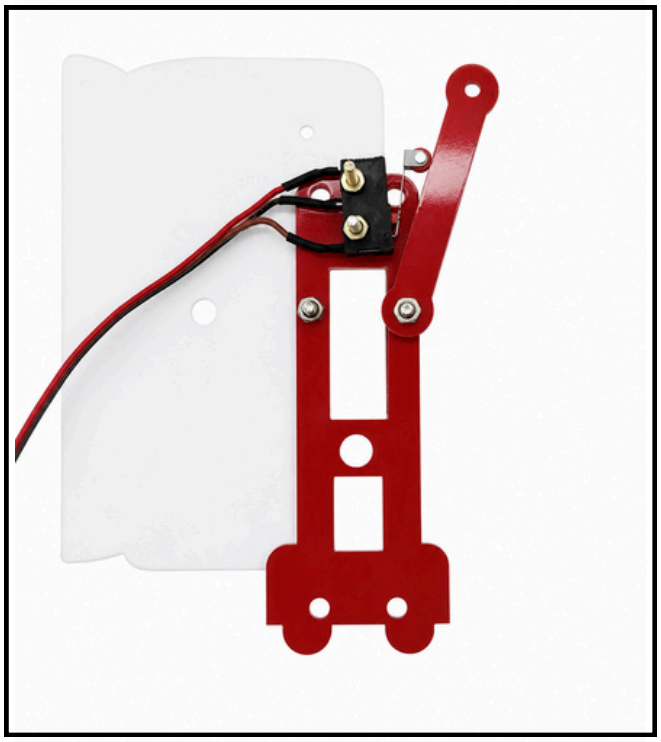


**Step 4:** Secure the L clamp as illustrated in the figure.



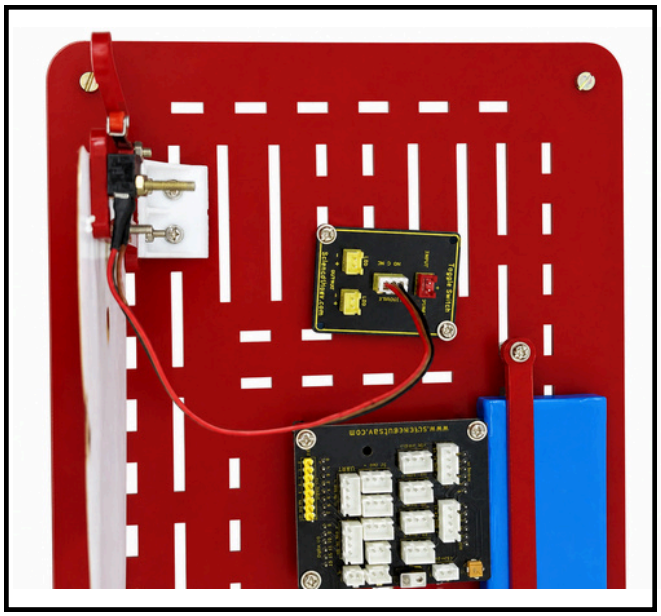
**Step 5:**

Here is a detailed description of how to connect the fridge template with acrylic pieces.

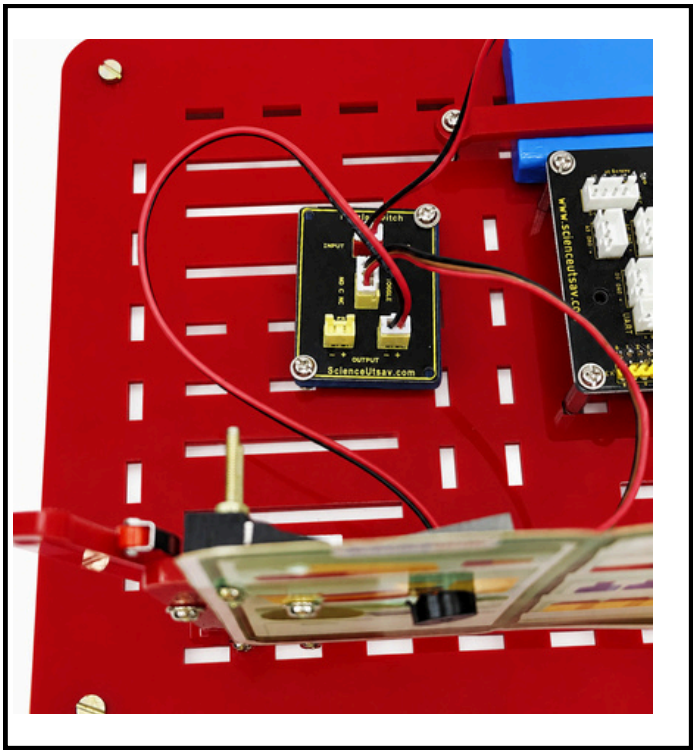


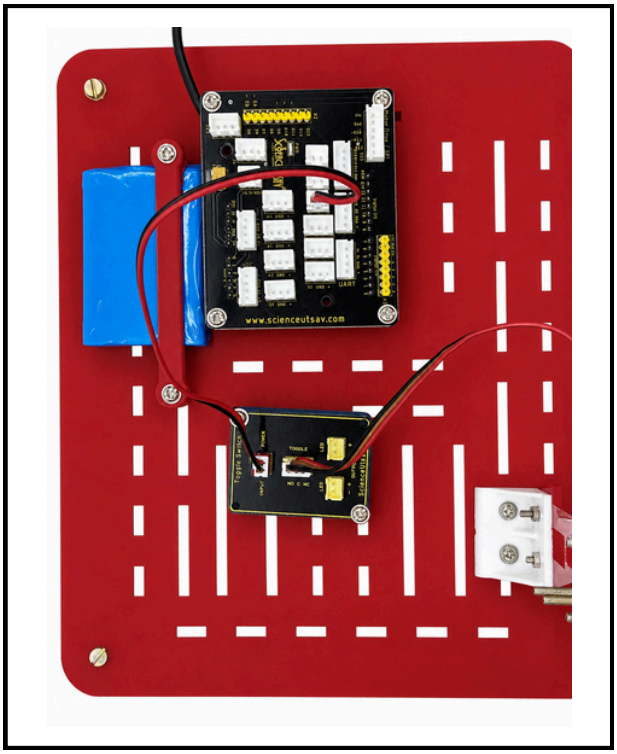






**Step 6:** Connect the buzzer to the toggle switch PCB as illustrated in the image.





**Step 7:** The model is now prepared. When you release the limit switch, it will begin to buzz.



# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

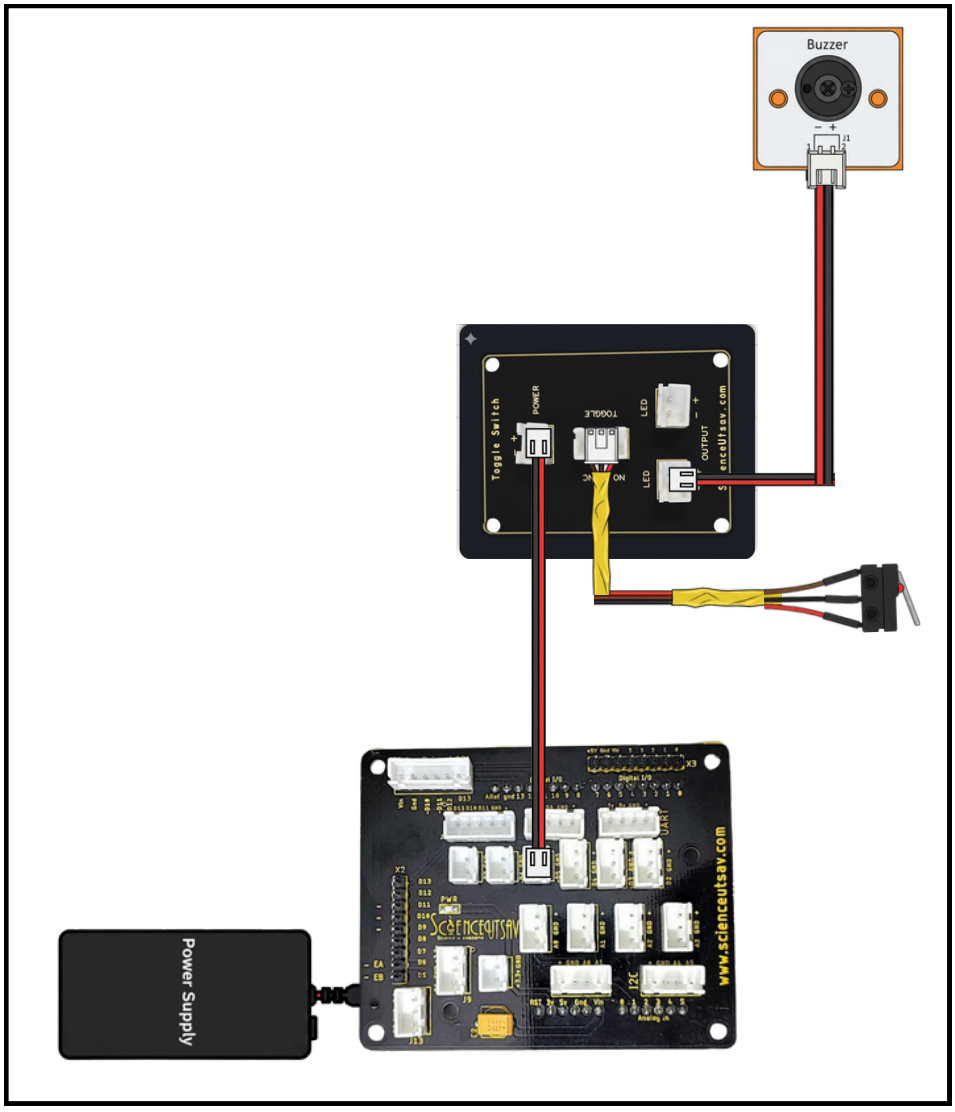
const int limitSwitch = 5;

void setup() {
  pinMode(limitSwitch, OUTPUT);
}

void loop() {
  digitalWrite(limitSwitch, HIGH);
}
```

# BEGINNER PROJECT

The circuit diagram of the project is given below

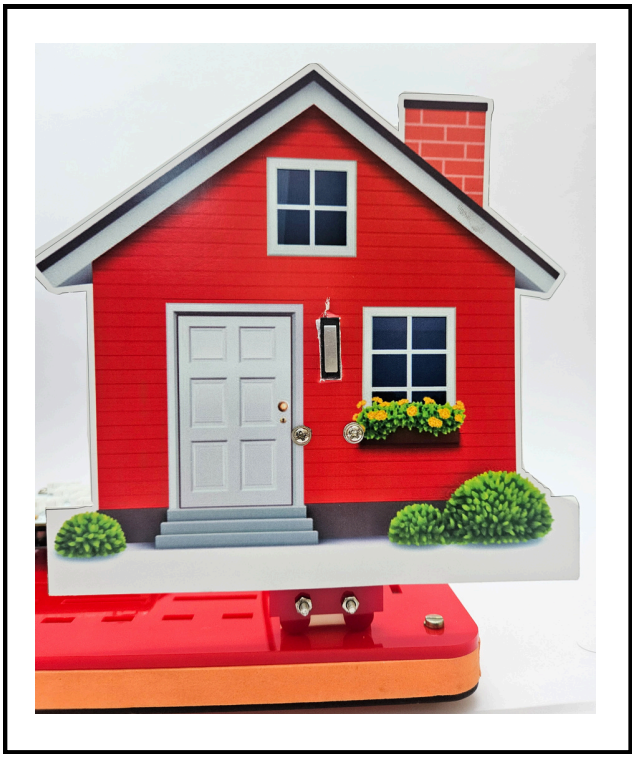



## CONNECTION:

1. TOGGLE SWITCH PCB IN D5
2. TOGGLE SWITCH PCB- LIMIT SWITCH, BUZZER

**Replace the previous template with the new one shown in the image to create the doorbell using the limit switch.**







# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

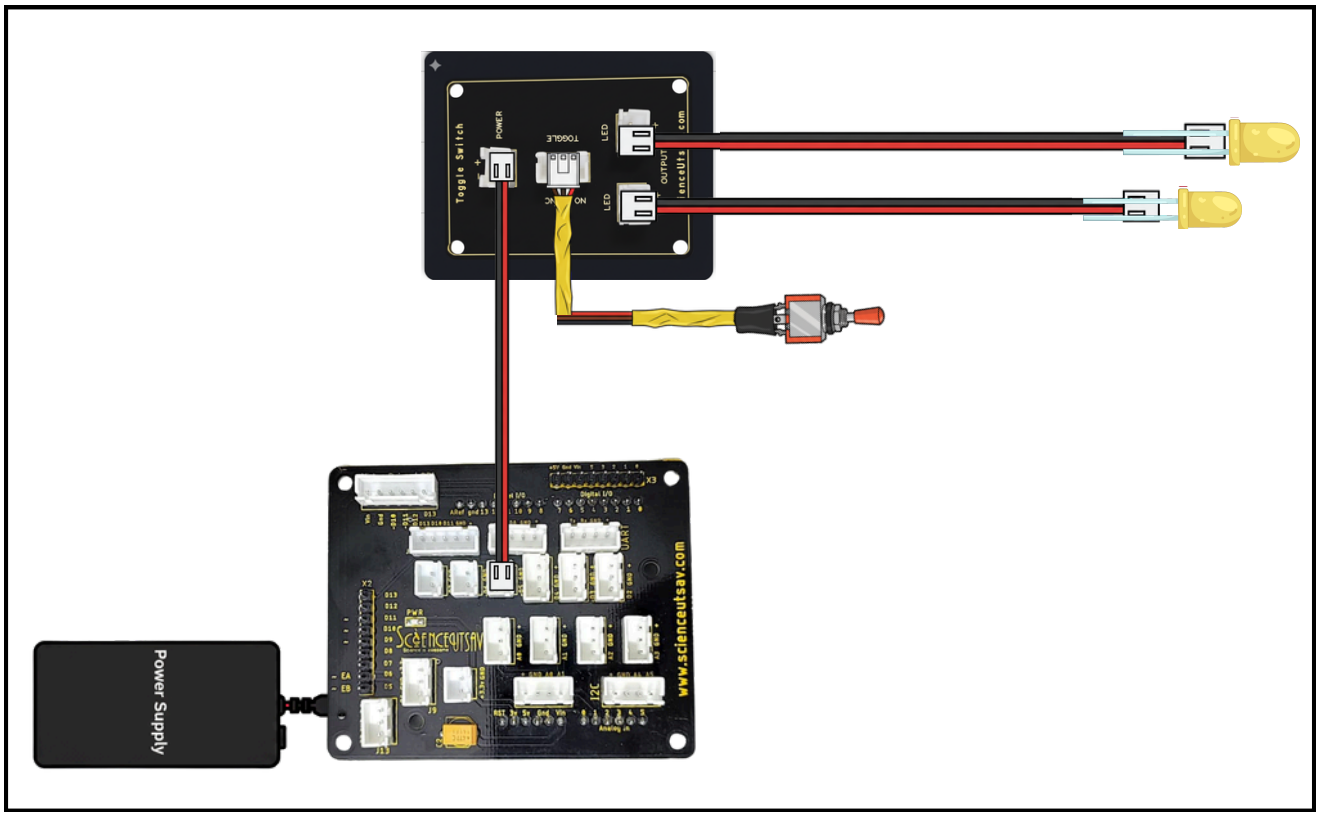
const int limitSwitch = 5;

void setup() {
  pinMode(limitSwitch, OUTPUT);
}

void loop() {
  digitalWrite(limitSwitch, HIGH);
  delay(500);
  digitalWrite(limitSwitch, LOW);
  delay(500);
}
```

# INTERMEDIATE PROJECT

The circuit diagram of the project is given below

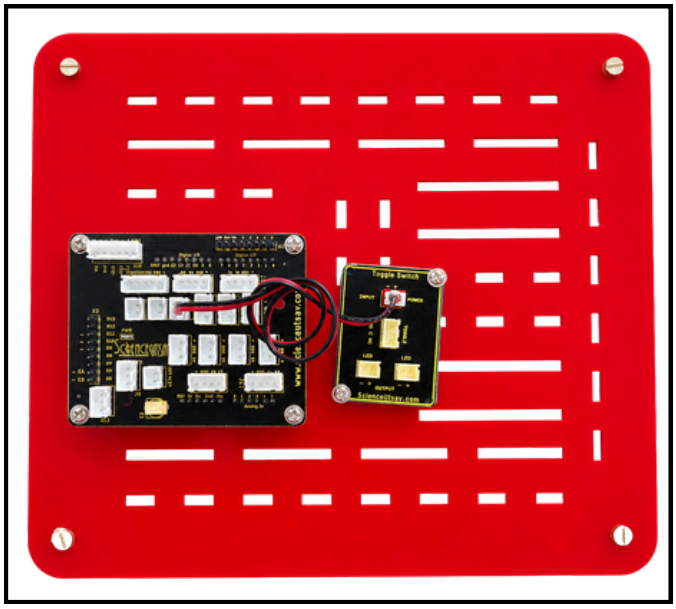


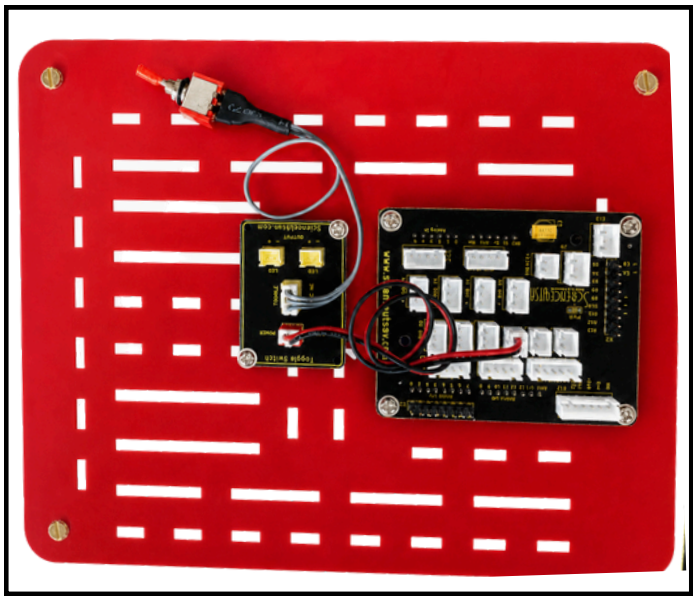

## CONNECTION:

1. TOGGLE SWITCH PCB IN D5
2. TOGGLE SWITCH PCB- LIMIT SWITCH, 2 DIP LEDs

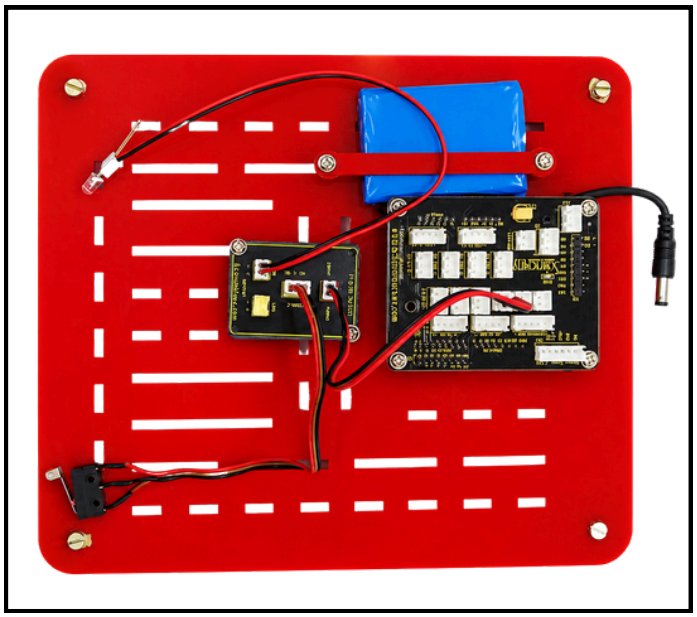
**Step 1:** Secure the Toggle Switch PCB to the Pegboard

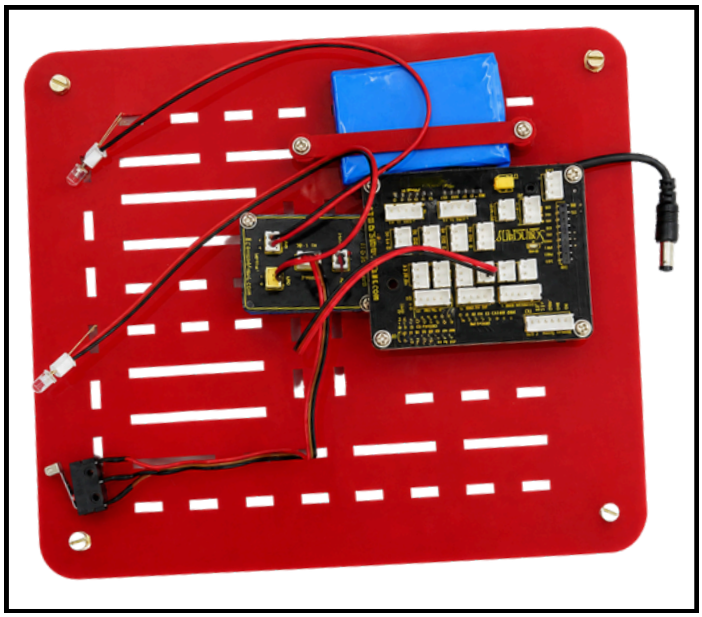
**Step 2:** Connect the 2-pin RMC to the D5 and the toggle switch PCB. Then, link the toggle switch to the toggle PCB.



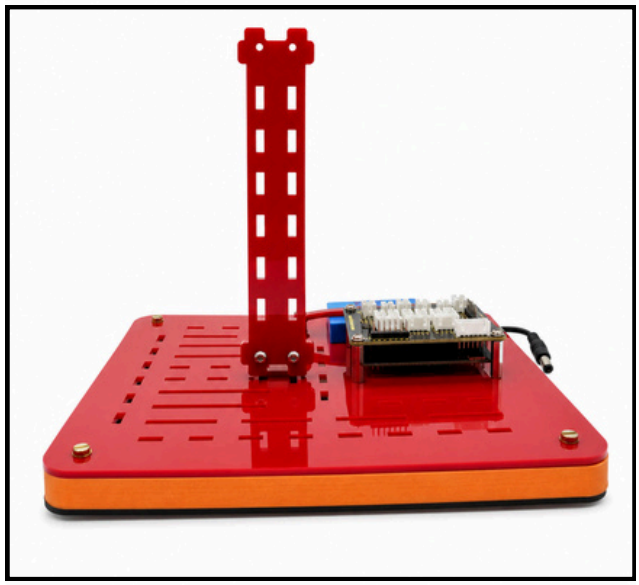


**Step 3:** Connect two DIP LEDs to the toggle switch PCB, as illustrated in the images below.





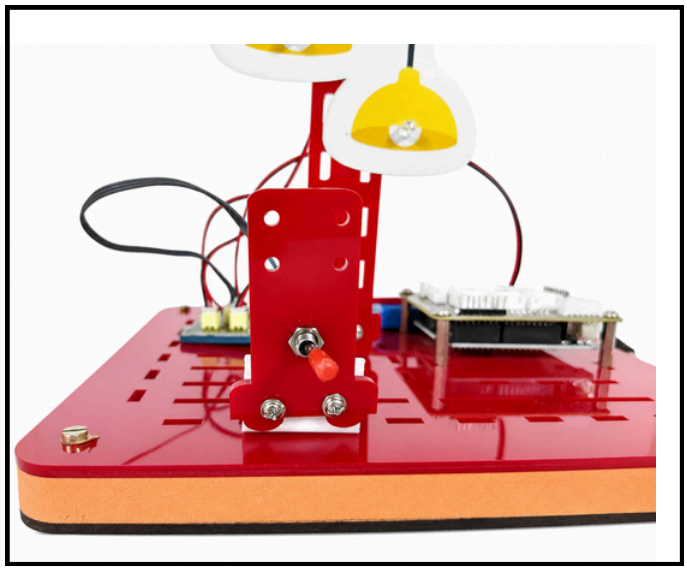
**Step 4:** Position the acrylic stand onto the pegboard using the L-clamp.



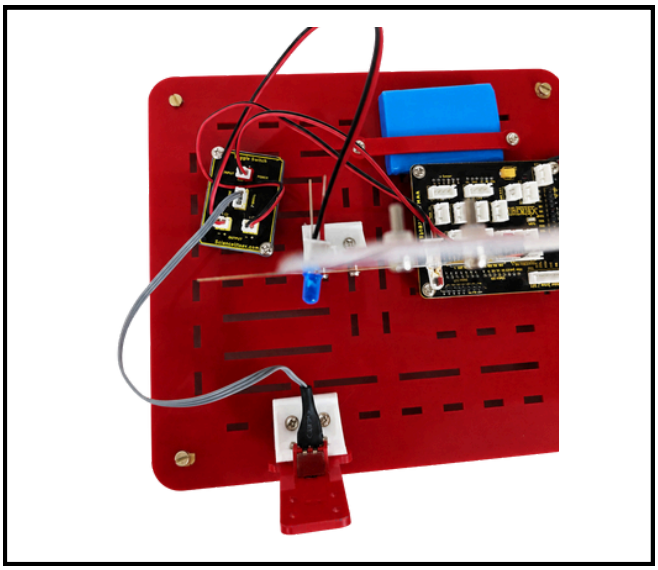
**Step 5:** Attach the template as illustrated in the image below.



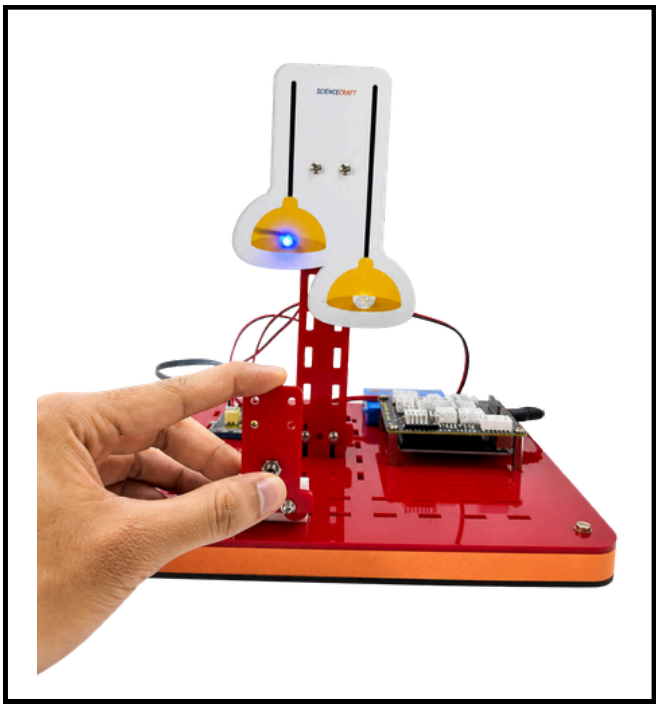
**Step 6:** Attach the toggle switch to the acrylic piece and secure it onto the pegboard using an L clamp.

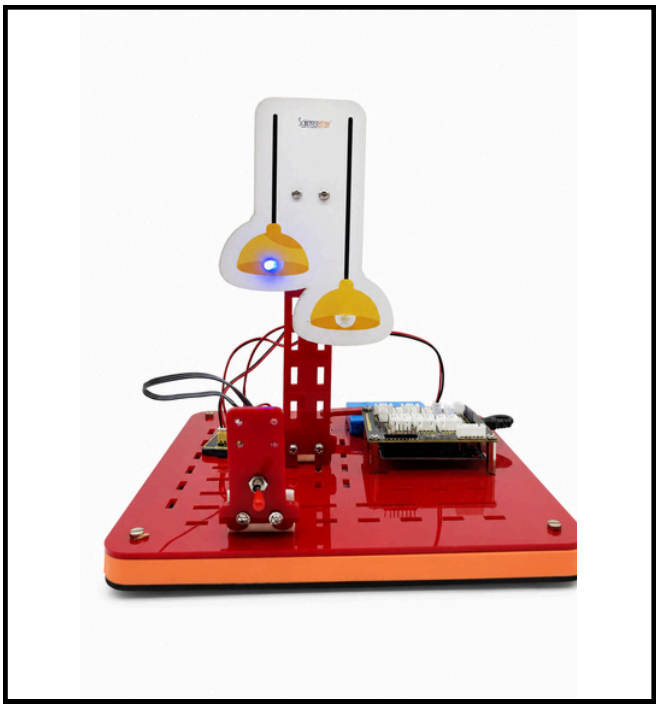


**Step 7:** Connect both LED dips to the light by linking the two-pin RMC to the toggle switch PCB.



**Hey, it’s working now! Operate the toggle switch to control both lights.**





# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

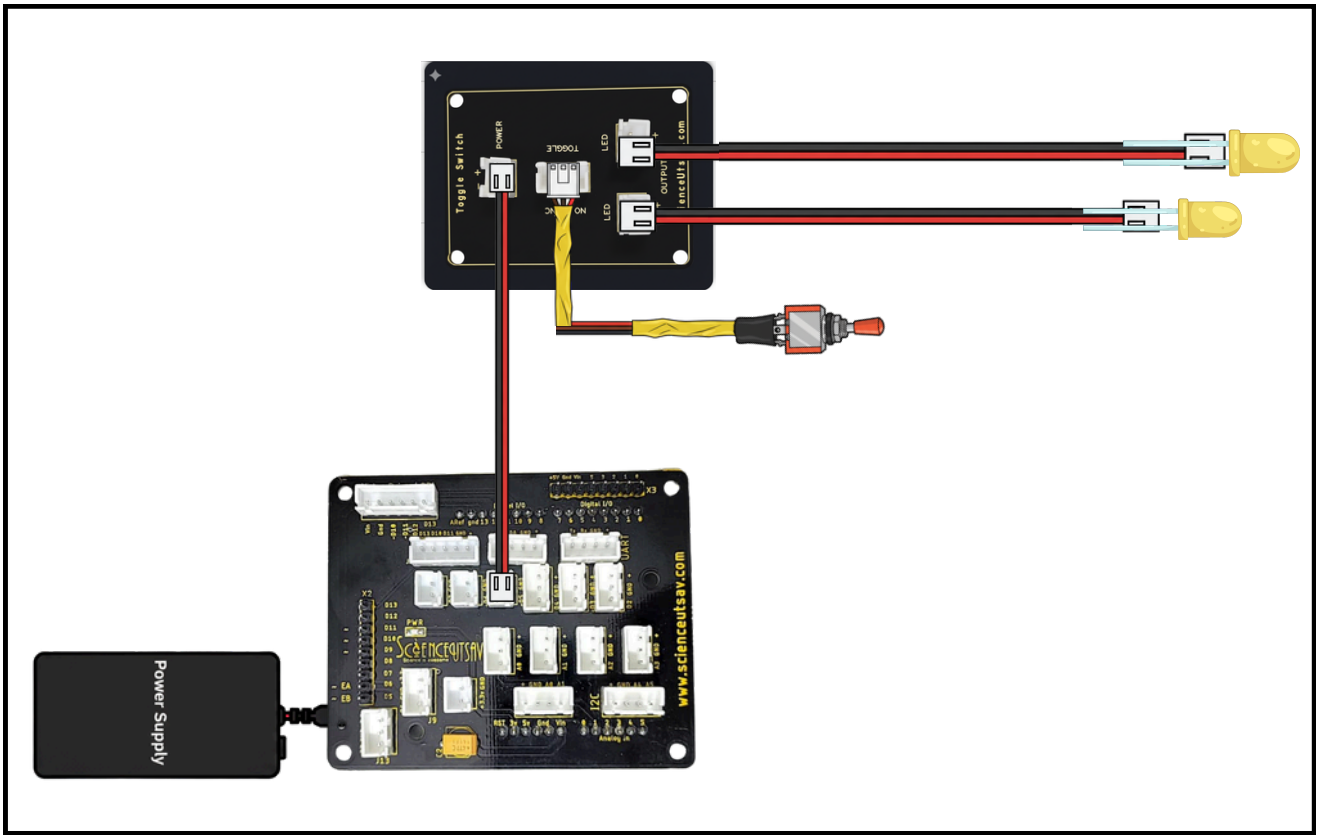
const int toggleSwitch = 5;

void setup() {
  pinMode(toggleSwitch, OUTPUT);
}

void loop() {
  digitalWrite(toggleSwitch, HIGH);
}
```

# INTERMEDIATE PROJECT

The circuit diagram of the project is given below




## CONNECTION:

1. TOGGLE SWITCH PCB IN D5
2. TOGGLE SWITCH PCB- LIMIT SWITCH, 2 DIP LEDs

**Follow the previous project steps and swap out the car template. Additionally, attach two red LEDs to the top of the car to serve as police lights.**



# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

const int toggleSwitch = 5;

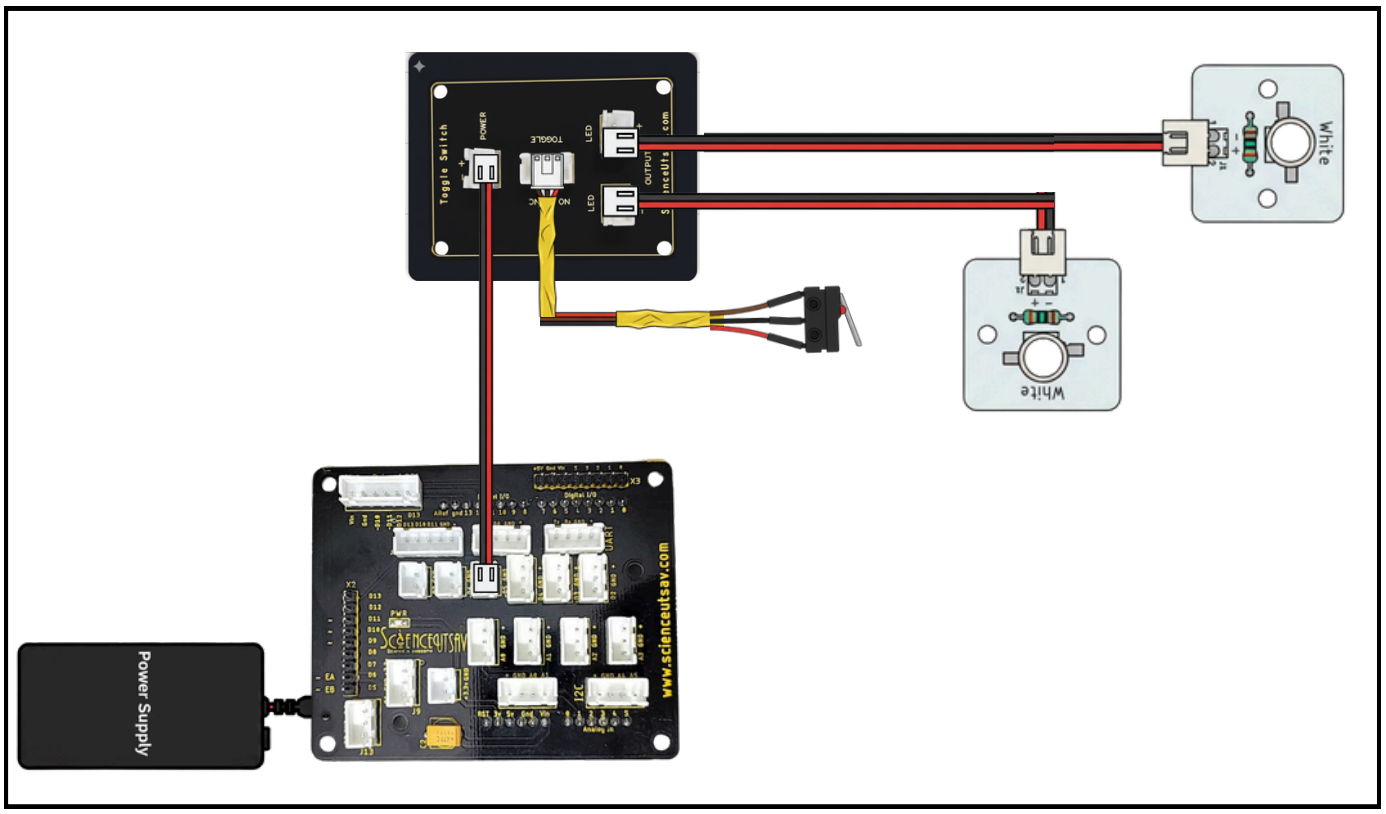
void setup() {
  pinMode(toggleSwitch, OUTPUT);
}

void loop() {
  digitalWrite(toggleSwitch, HIGH);

}
```

# ADVANCE PROJECT

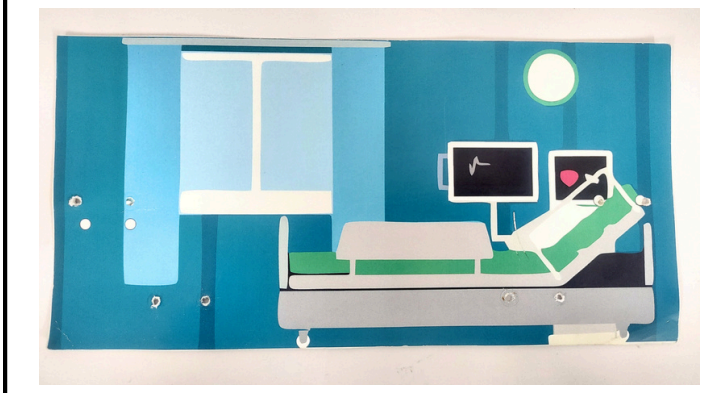
The circuit diagram of the project is given below

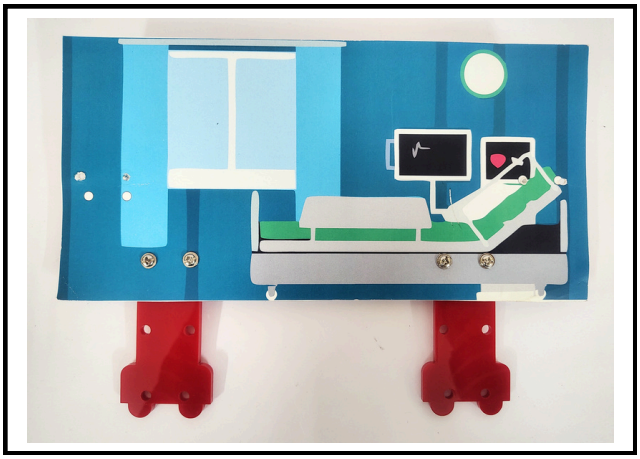



## CONNECTION:

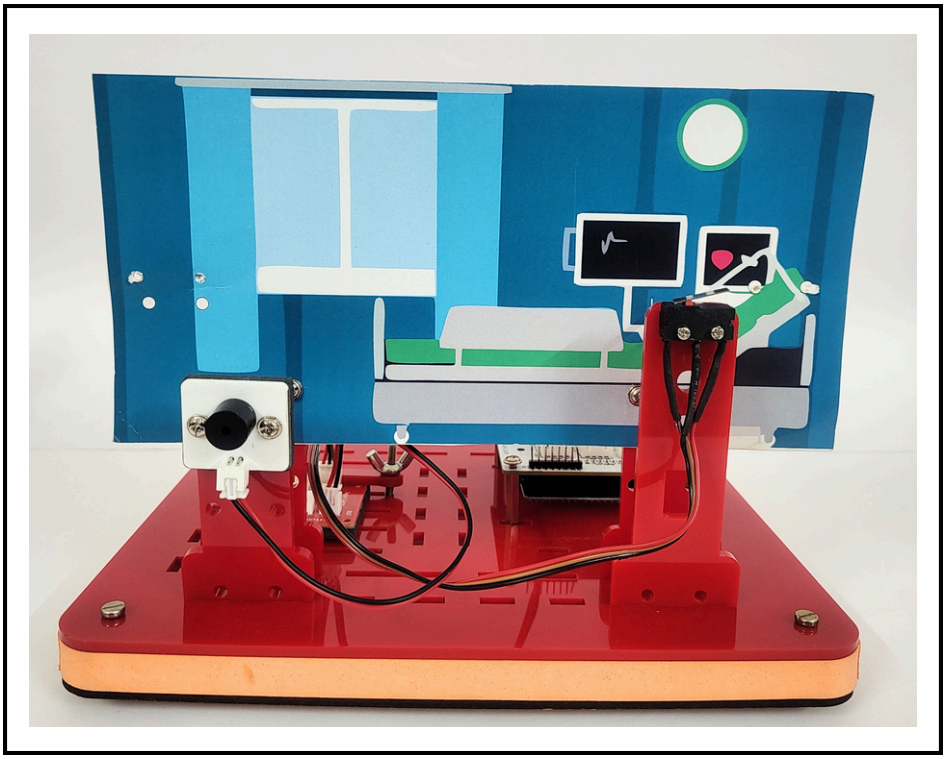
1. TOGGLE SWITCH PCB IN D5
2. TOGGLE SWITCH PCB- LIMIT SWITCH, 2 1W LEDs

**Step 1**: Adjust the Template Stands
Begin by taking this template and modifying the stands according to the examples illustrated in the images below.





**Step 2:** Connect the Buzzer and Limit Switch
Follow the instructions in the image to attach the buzzer and the limit switch. Once the limit switch is pressed, it will activate the siren as an emergency alert.



# code

```
/*
Simple Arduino Code
Pin logic based on updated component list
and custom switch PCB logic.
*/

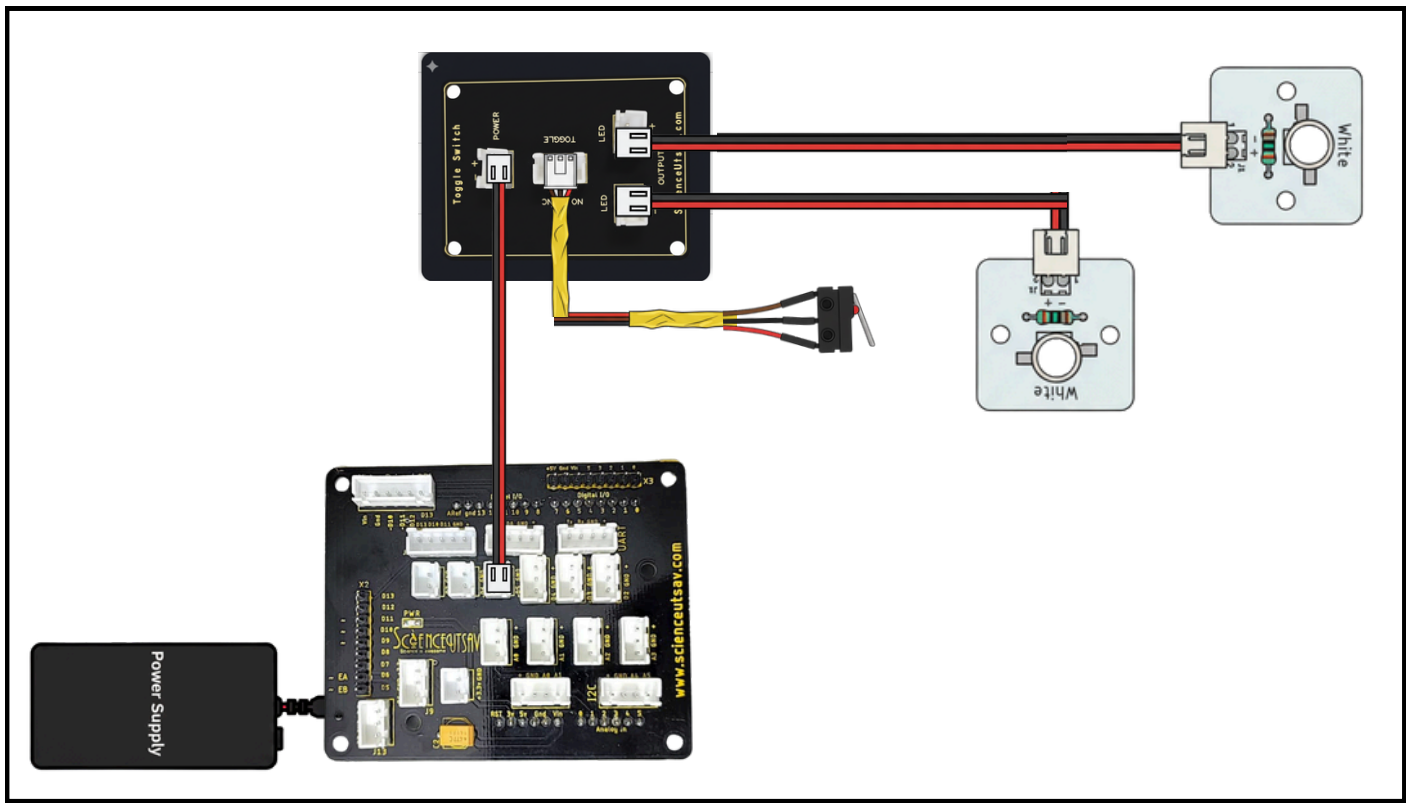
const int twoWaySwitch = 5;

void setup() {
  pinMode(twoWaySwitch, OUTPUT);
}

void loop() {
  digitalWrite(twoWaySwitch, HIGH);
}
```

# ADVANCE PROJECT

The circuit diagram of the project is given below




## CONNECTION:

1. TOGGLE SWITCH PCB IN D5
2. TOGGLE SWITCH PCB- LIMIT SWITCH, 2 1W LEDs

**Follow the steps from the previous session and substitute the template with the one provided below. Additionally, incorporate the two DIP LEDs as indicators.**



# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

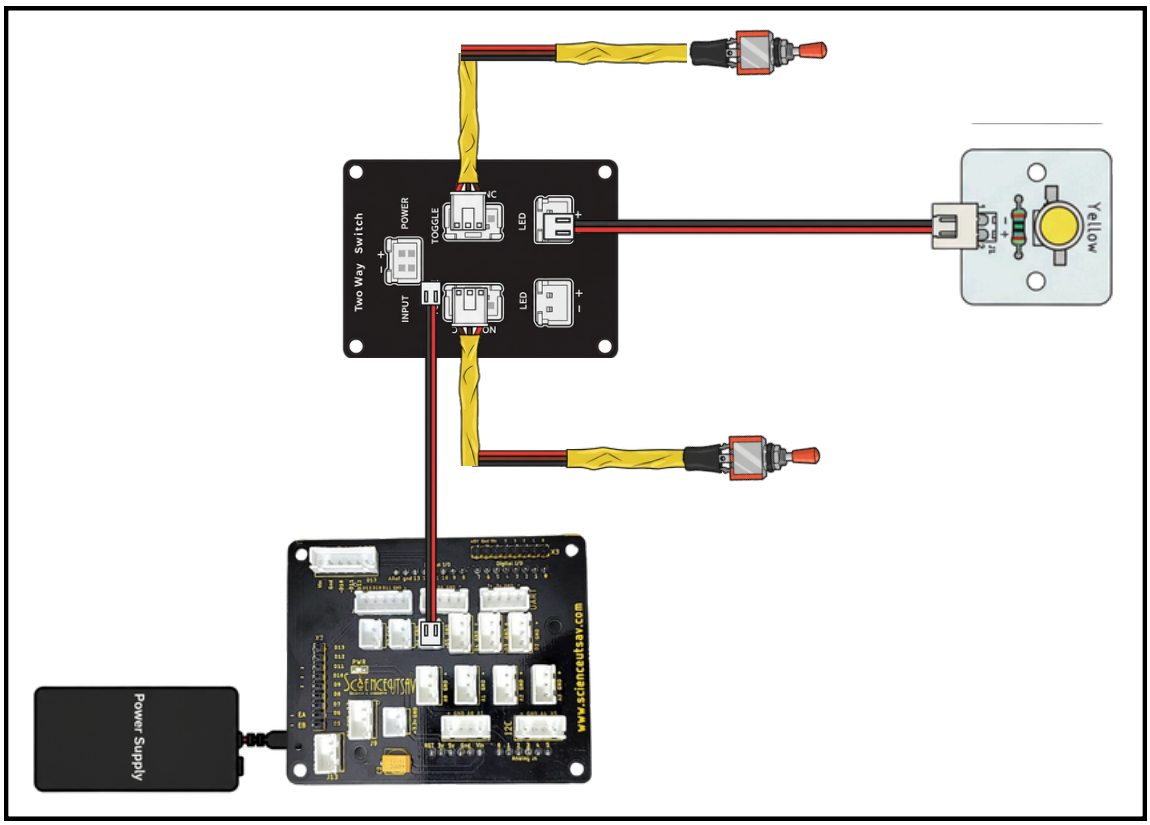
const int twoWaySwitch = 5;

void setup() {
  pinMode(twoWaySwitch, OUTPUT);
}

void loop() {
  digitalWrite(twoWaySwitch, HIGH);
  delay(500);
  digitalWrite(twoWaySwitch, LOW);
  delay(500);
}
```

# EXPERT PROJECT

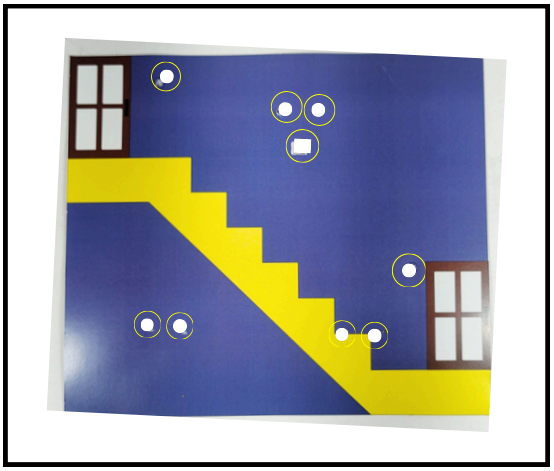
The circuit diagram of the project is given below

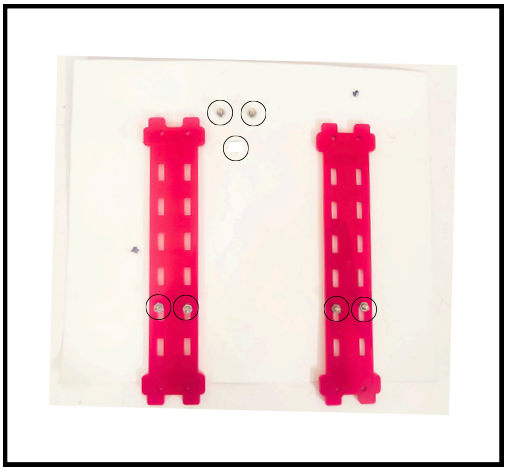



## CONNECTION:

1. TWO-WAY SWITCH PCB IN D5
2. TOGGLE SWITCH PCB- 2 TOGGLE SWITCHES, 1W LED

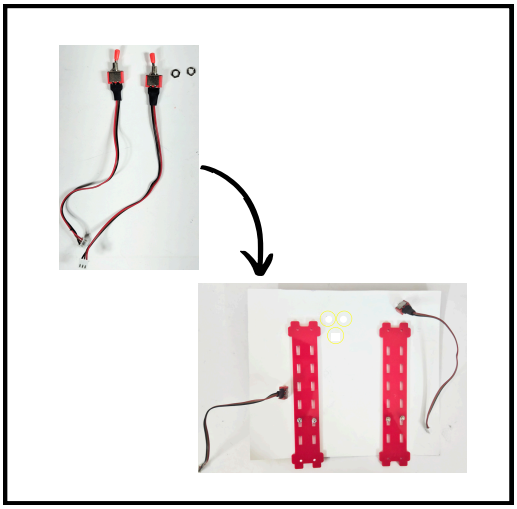
**Step 1:** Utilize the template provided below, and incorporate the acrylic stands as illustrated in the images.

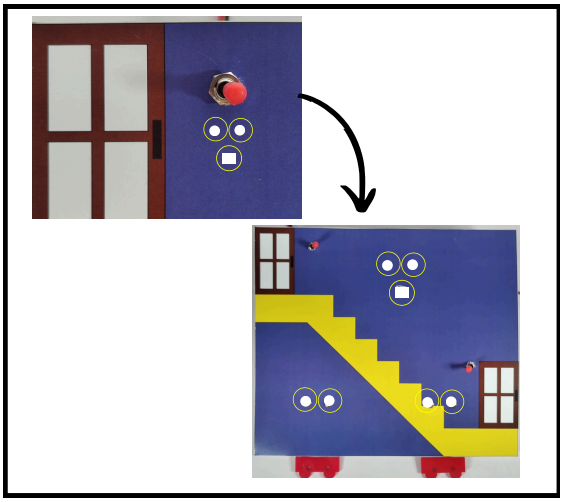




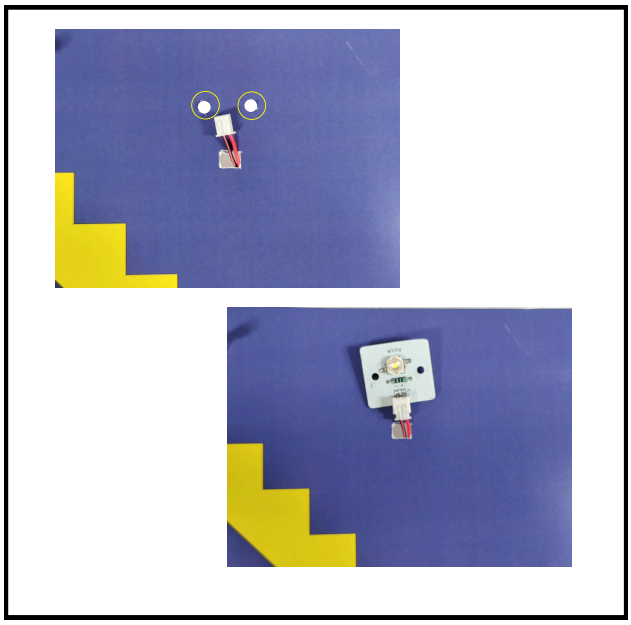
**Step 2:** Now, incorporate two toggle switches into the templates, as illustrated in the images below.

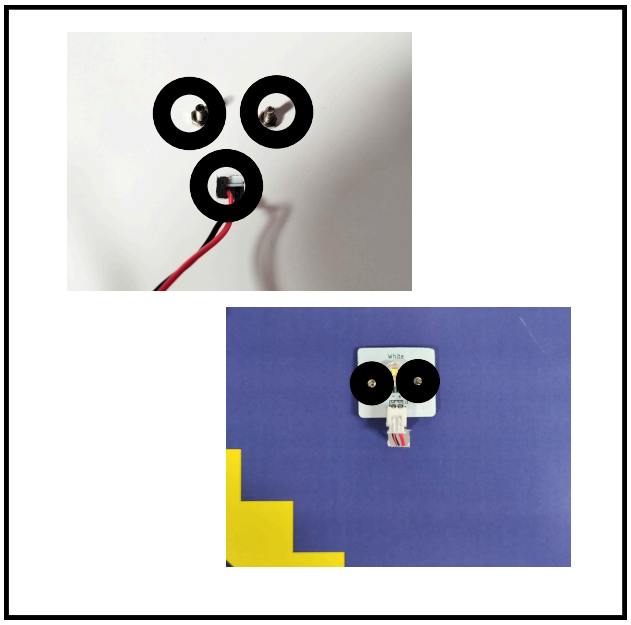




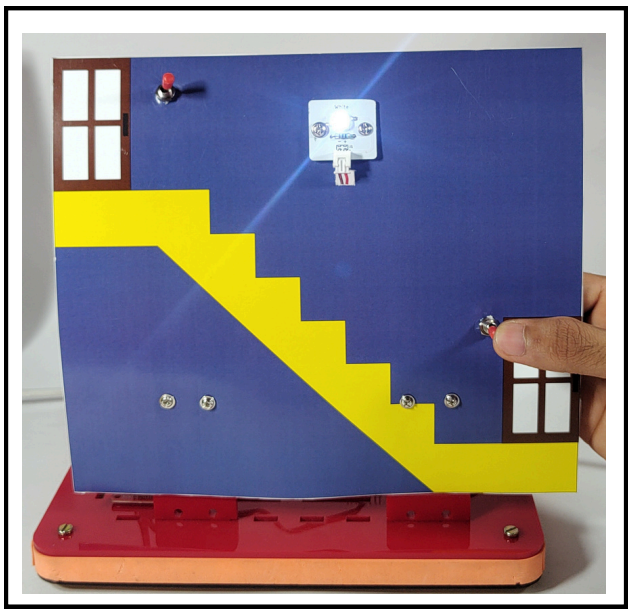


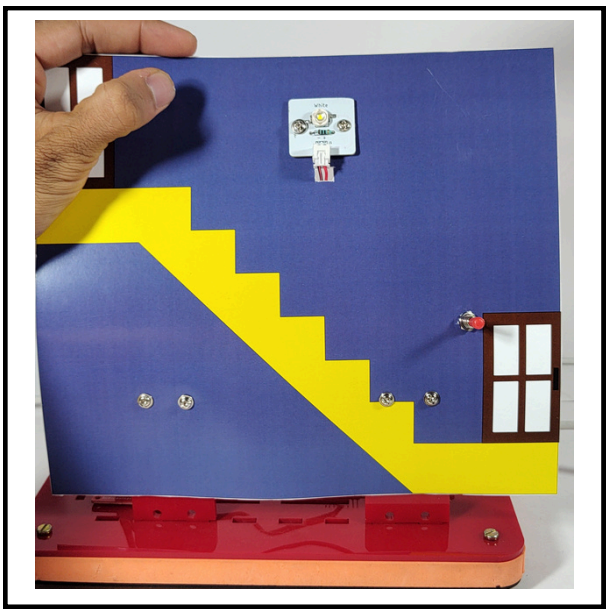
**Step 3:** Adding the 1-Watt LED to the Template
Now, incorporate the 1-watt LED into the template by carefully following the precise steps outlined below.





**Step 4:** Now that the two-way switch system is ready, you can observe that the 1-watt LED is being controlled by the two switches. This setup allows operation from either switch.





# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

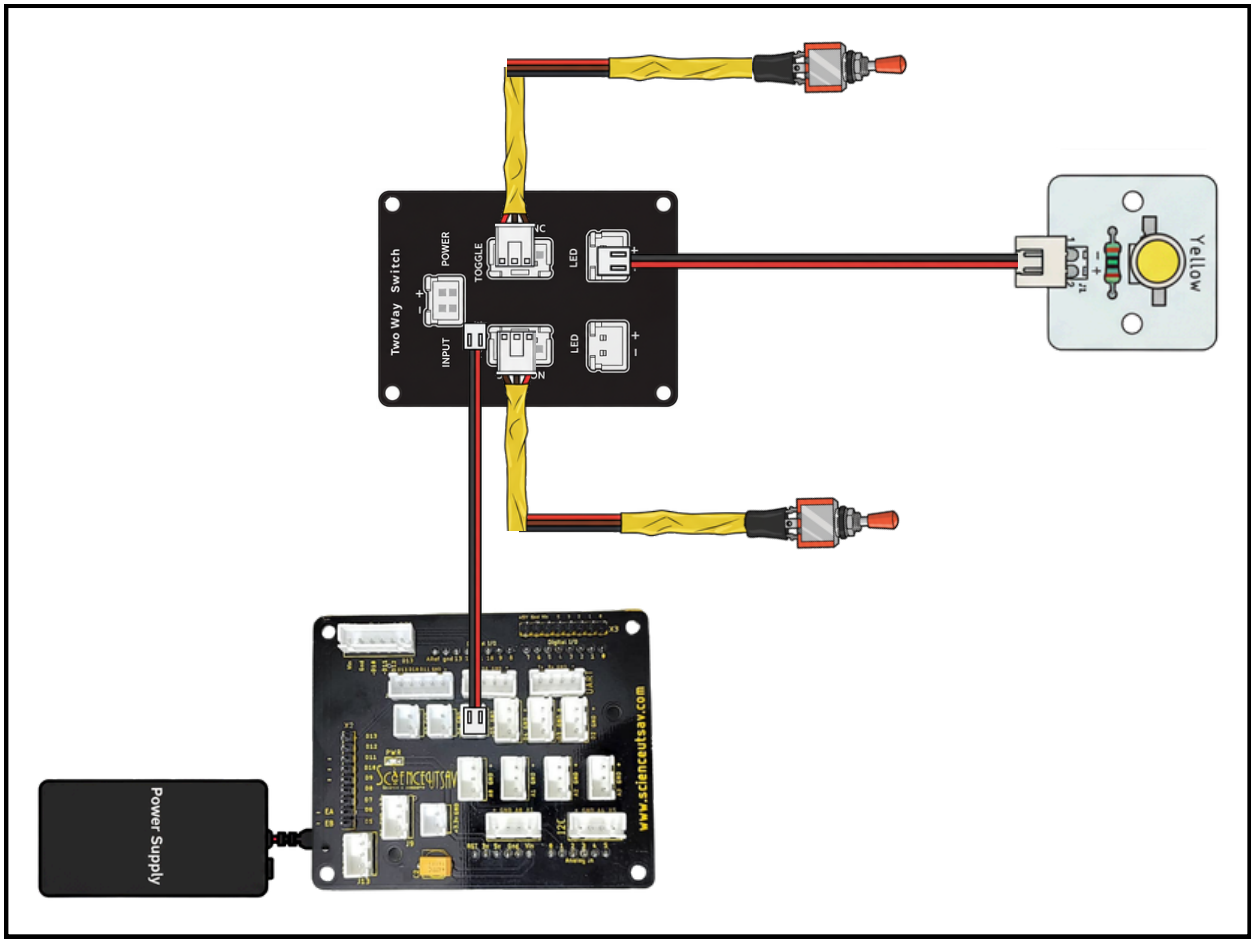
const int twoWaySwitch = 5;

void setup() {
  pinMode(twoWaySwitch, OUTPUT);
}

void loop() {
  digitalWrite(twoWaySwitch, HIGH);
}
```

# EXPERT PROJECT

The circuit diagram of the project is given below




## CONNECTION:

1. TWO-WAY SWITCH PCB IN D5
2. TOGGLE SWITCH PCB- 2 TOGGLE SWITCHES, 1W LED

Now, follow the exact steps from the previous session and simply replace the template below by attaching the two toggle switches to the right and left side walls. Position the I Watt LED towards the bedside, allowing you to control the bed light through the two switches.



# code

```
/*
Simple Arduino Code
Pin logic based on updated component list and
custom switch PCB logic.
*/

const int toggleSwitch = 5;

void setup() {
  pinMode(toggleSwitch, OUTPUT);
}

void loop() {
  digitalWrite(toggleSwitch, HIGH);
}
```